

As a Senior Database Architect, I have designed the schema for the AI-Powered Hospital Management System. This design prioritizes HIPAA compliance, data integrity, and the high-concurrency requirements of a clinical environment.

1. Database Overview

The system uses a **Relational Database Management System (RDBMS)** to ensure ACID compliance. The architecture is modular, separating identity management, clinical records, resource scheduling, and AI-audit logs to ensure the security and scalability required for 99.9% uptime.

2. ER Diagram Description

- **Users/Roles:** Centralized Auth for all stakeholders.
- **Patient Records:** Core entity linked to appointments, EHR, and Billing.
- **Clinical/AI:** EHR tables store actual records, while `AI\_Diagnostic\_Logs` stores suggestions and the required human-in-the-loop signatures.
- **Operational:** Resources (beds/rooms) track current occupancy for AI-driven staffing.

3. Tables & Schema

Core Schema (SQL)

```
-- Role-based access control
CREATE TABLE Roles (role_id SERIAL PRIMARY KEY, role_name VARCHAR(50));

-- Users (Staff & Admin)
CREATE TABLE Users (
    user_id UUID PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(150) UNIQUE,
    role_id INT REFERENCES Roles(role_id),
    password_hash TEXT
);

-- Patients
CREATE TABLE Patients (
    patient_id UUID PRIMARY KEY,
    full_name VARCHAR(100),
    dob DATE,
    contact_info TEXT,
    consent_for_ai BOOLEAN DEFAULT FALSE -- Business Rule requirement
);

-- Appointments
```

```

CREATE TABLE Appointments (
    appt_id UUID PRIMARY KEY,
    patient_id UUID REFERENCES Patients(patient_id),
    doctor_id UUID REFERENCES Users(user_id),
    appt_time TIMESTAMP,
    triage_level VARCHAR(20), -- AI-Assisted Triage output
    status VARCHAR(20)
);

-- EHR Records
CREATE TABLE EHR_Records (
    record_id UUID PRIMARY KEY,
    patient_id UUID REFERENCES Patients(patient_id),
    physician_id UUID REFERENCES Users(user_id),
    diagnosis TEXT,
    treatment_plan TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- AI Insights & Human-in-the-loop audit
CREATE TABLE AI_Audit_Log (
    audit_id UUID PRIMARY KEY,
    record_id UUID REFERENCES EHR_Records(record_id),
    ai_suggestion TEXT,
    clinician_verified_at TIMESTAMP,
    clinician_id UUID REFERENCES Users(user_id) -- Ensures accountability
);

-- Resource Management (Beds)
CREATE TABLE Hospital_Resources (
    resource_id SERIAL PRIMARY KEY,
    type VARCHAR(50), -- e.g., 'Bed', 'Ventilator'
    is_occupied BOOLEAN DEFAULT FALSE
);

```

4. Relationships

- **One-to-Many:** `Patients` to `Appointments`, `Patients` to `EHR\_Records`.
- **One-to-Many:** `Users` (Doctors) to `Appointments` and `EHR\_Records`.
- **One-to-One:** `AI\_Audit\_Log` to `EHR\_Records` (for tracking verification).

5. Indexes

- **Performance (NFR4):**
- ```

* `CREATE INDEX idx_appt_time ON Appointments(appt_time);` (Fast retrieval for daily dashboards).
* `CREATE INDEX idx_patient_lookup ON Patients(full_name);`
* `CREATE INDEX idx_ehr_patient ON EHR_Records(patient_id);` (Optimization for patient record retrieval).

```

## 6. Constraints

- **Data Integrity:** All foreign keys use `ON DELETE RESTRICT` to prevent accidental loss of clinical history.
- **Security (NFR1):** Use of UUIDs for primary keys to prevent enumeration attacks.
- **Business Rule:** `CHECK (consent_for_ai IS NOT NULL)` ensures data usage compliance.

## 7. Normalization Notes

- **3NF (Third Normal Form):** All tables are normalized to 3NF. We avoid update anomalies by keeping user credentials separate from medical observations.
- **Auditability:** The `AI_Audit_Log` acts as a specialized table to maintain an immutable record of AI interaction, separating analytical data from transactional EHR records, preventing database bloat while maintaining strict compliance.

## 8. Compliance & Security

- **Encryption:** The application layer must handle transparent data encryption (TDE) for PHI fields (e.g., `diagnosis`, `contact_info`).
- **Audit Logging:** Every insert/update on `EHR_Records` should be mirrored in a system-wide Audit Log table triggered by database triggers.